



OBJECT DETECTION RADAR SYSTEM

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TEAM 13

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1. Introduction

Modern autonomous systems are the result of decades of rapid progress in robotics, control systems and automation. As their widespread use continues to develop, humans are relying increasingly on these systems to perform important tasks. System reliability is critical to guaranteeing the viability of any autonomous vehicle; consequentially, the demand for support for these systems will surge as their implementation becomes commonplace.

1.1 Objectives

The ODRS' mission is to increase the viability of autonomous systems by providing them with a reliable method for vehicle guidance and navigation.

Primary Objectives:

- **Object Detection:**
Implement ultrasonic sensors capable of detecting both static and dynamic objects within a defined field of view. The sensors are integrated with a microcontroller to continuously monitor for changes in distance, enabling real-time object presence recognition.
- **User-Tuned Range Control:**
Incorporate an RFID-based user input interface to dynamically adjust the scanning range and angle of each sensor. This customization enhances adaptability to various environments by modifying detection parameters on-the-fly.
- **360° Scanning Capability:**
Utilize a continuous rotation servo motor at the base of the system to achieve panoramic environmental scanning. This allows the sensor array to rotate indefinitely, enabling complete 360-degree spatial awareness without mechanical limits.

Secondary Objectives:

- **Velocity Detection:**
Estimate the velocity of moving objects by calculating the rate of change in distance measurements over time. This is achieved through periodic sampling of ultrasonic sensor data and applying basic kinematic equations.
- **Object Range Estimation:**
Calculate the real-time distance between the sensor and detected objects using time-of-flight data from ultrasonic pulses. The system processes this data to provide accurate range information within the configured detection threshold.

1.2 Functionalities

Primary Features:

- Continuous Servo Rotation – The centrally positioned continuous rotation servo functions as the primary actuator for the system’s 360° scanning capability. It enables uninterrupted, bidirectional rotation, allowing the sensor platform to perform real-time panoramic sweeps based on user-defined parameters.
- User-input Vertical Alignment - Two top-mounted SG90 servo motors enable independent angular adjustment of the ultrasonic sensors. These servos allow user-defined orientation within a $\pm 45^\circ$ range, providing customizable vertical alignment for enhanced flexibility in object detection coverage.
- Range Toggling - Allows dynamic adjustment of the sensor detection range, enabling the system to modify its object scanning threshold based on input parameters.

Secondary Features:

- LED lights for user feedback - The sensor housing was designed in CAD with integrated mounting points for red and green LEDs, serving as visual indicators to convey real-time object detection status to the user.
- Sliding Door - A sliding access panel was incorporated into the mechanical design to facilitate streamlined troubleshooting, maintenance, and component access for both users and developers, while also contributing to the system’s overall aesthetic integration.

1.3 Methodology

With any project, comes an extensive design process. The ODRS was no exception to this rule – we went through many different stages of designs before ultimately deciding on our final design. Our first constraints were set loosely and were based off the Keypad Matrix’s dimensions of 7.5cm in height. We wanted to make the ODRS as small as possible, so we used the exact height of the keypad to minimize unneeded space.

Initial Design Approach:

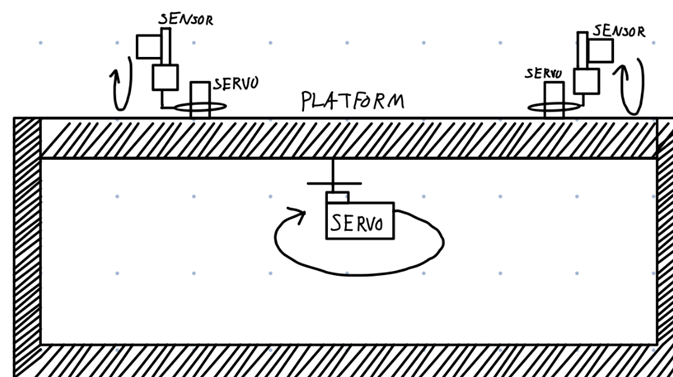


Figure 1: Initial Design Concept Side View

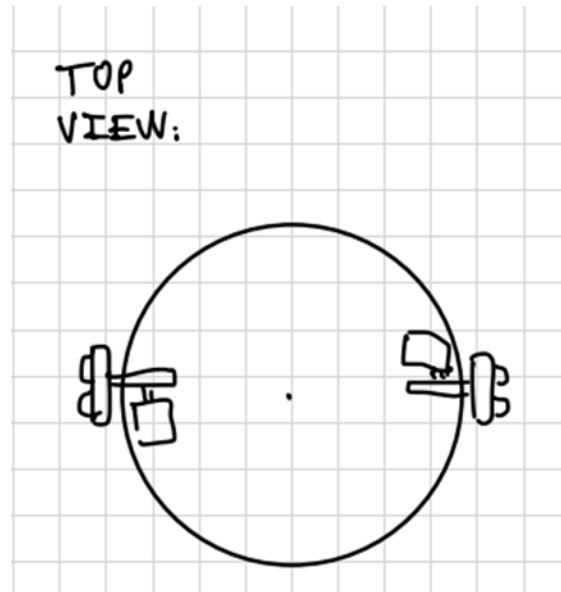


Figure 2: Initial Design Concept Top View

The system features a two-layer cylindrical architecture optimized for full-range environmental scanning. A 360° continuous rotation servo enables omnidirectional motion, while vertically mounted customizable servos provide adjustable sensor orientation. The lower layer integrates the primary PCB, rotation mechanism, and base components, while the upper layer houses the top PCB, sensors, and power supply. This configuration maximizes viewing coverage, minimizes blind spots, and ensures efficient spatial data acquisition within a compact and modular design.

Final Design Approach:

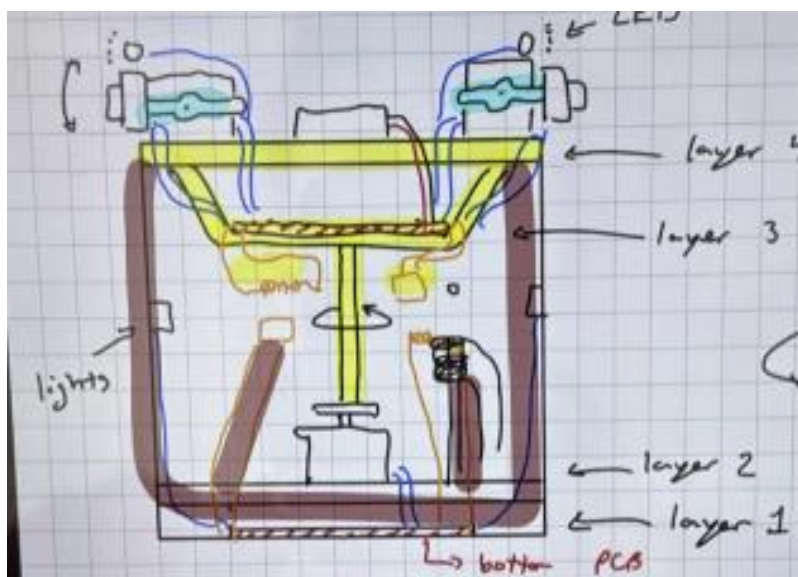


Figure 3: Final Design Concept Interior View

It comprises four structural layers:

- Layer 1 houses the bottom PCB,
- Layer 2 contains the bottom iron core, bottom hall effect sensor, and 360° servo motor
- Layer 3 supports the top PCB, top iron core, and top hall effect sensor
- Layer 4 integrates the vertical alignment servos, ultrasonic sensors, and batter pack

Prototyping

➤ 1st Design:

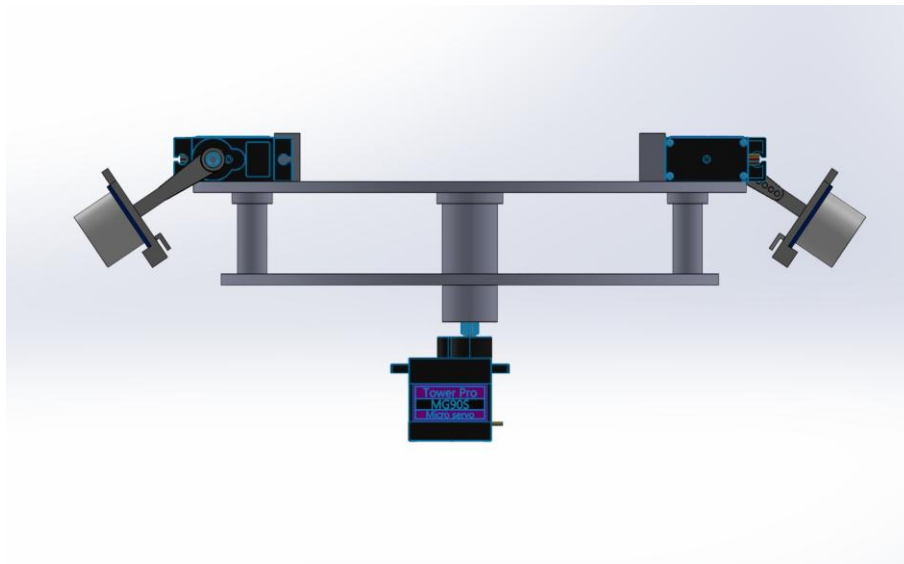


Figure 4: Initial Prototype

Integrated from the initial sketches, features a compact cylindrical two-layer architecture designed for full 360° rotational coverage. The lower layer integrates a continuous rotation servo and accommodates a breadboard or PCB for core control circuitry. The upper layer supports two vertically mounted actuating servos with attached sensors, enabling customizable directional scanning. While provisioned for modular expansion, the current implementation excludes RFID functionality. The design prioritizes wide-angle sensing and scalable integration within a streamlined structural footprint.

➤ 2nd Design:

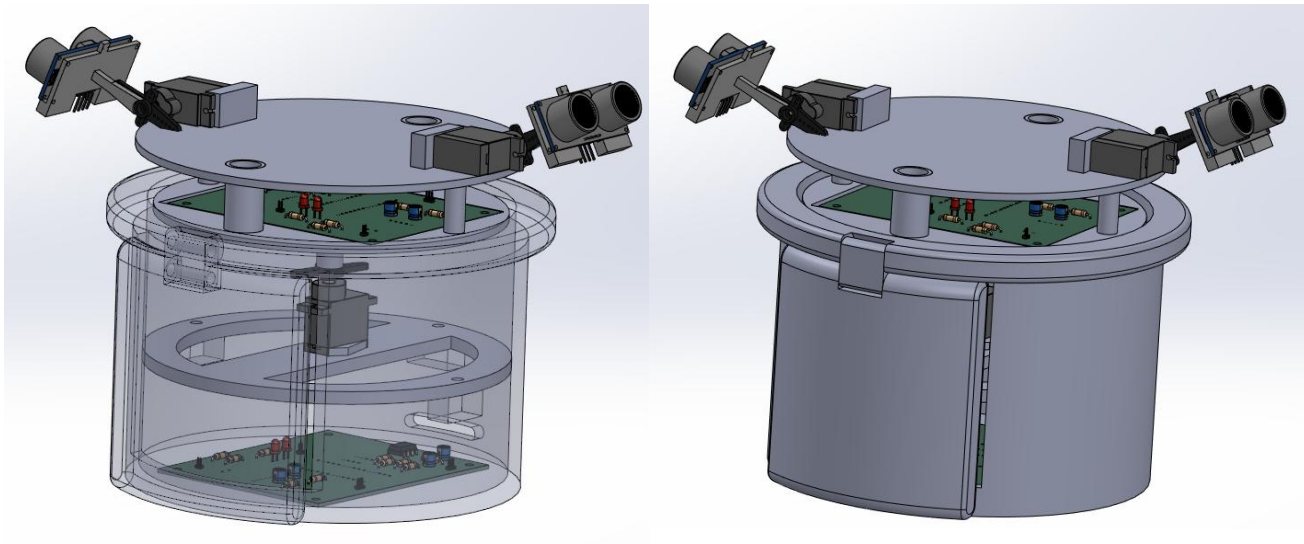


Figure 5: Second Prototype

Integrating from the final design sketch, the second design prototype utilizes a four-layer cylindrical structure with an integrated access door for enhanced modularity and maintenance. However, the latched door would not be viable to produce with the 3D printer, so a sliding door was implemented instead to guarantee printing success. A hole in the door was also added for USB access.

➤ 3rd (Final) Design:

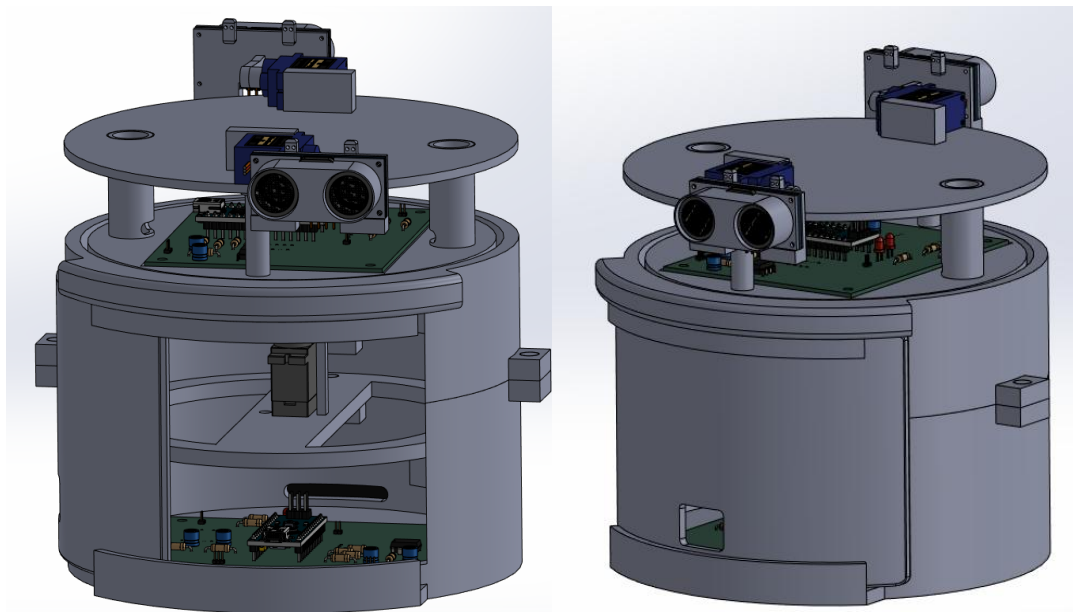


Figure 6: Final Prototype

Including everything, the final prototype features a four-layer cylindrical architecture with a focus on modularity, rotational coverage, and mechanical accessibility. Due to 3D printing limitations, the base structure was split into two segments, and a redesigned access door was implemented to improve assembly and internal access. All structural components were fabricated using Ahmed's 3D printer, with adjusted tolerances to ensure mechanical stability.

- Layer 1: Houses the bottom PCB for control circuitry and power distribution.
- Layer 2: Contains the 360° continuous rotation servo, bottom iron core, and a Hall effect sensor for rotational tracking.
- Layer 3: Supports the top PCB, top iron core, and a secondary Hall effect sensor for rotational alignment.
- Layer 4: Mounts two vertical servos with ultrasonic sensors, along with the battery pack for power autonomy.

The integrated swing-door mechanism allows tool-free access for maintenance and prototyping flexibility. Mechanical design considerations—including door integration, base segmentation, and structural reinforcement—will be further evaluated in the Mechanical Module.

➤ PCB Creation:

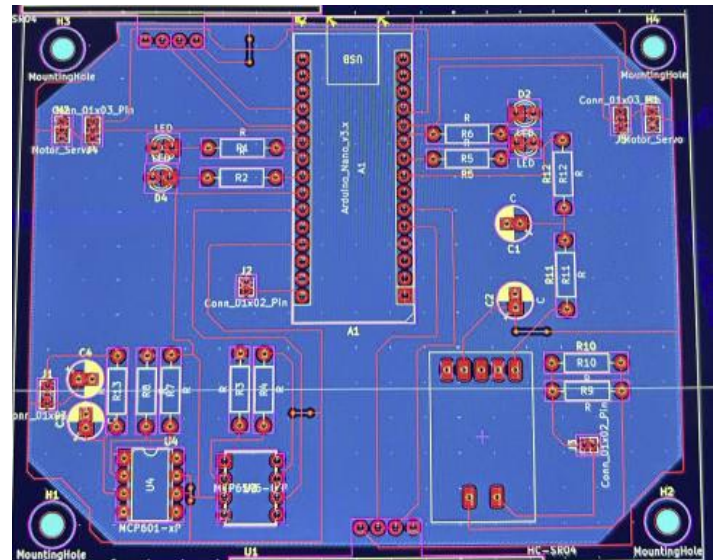
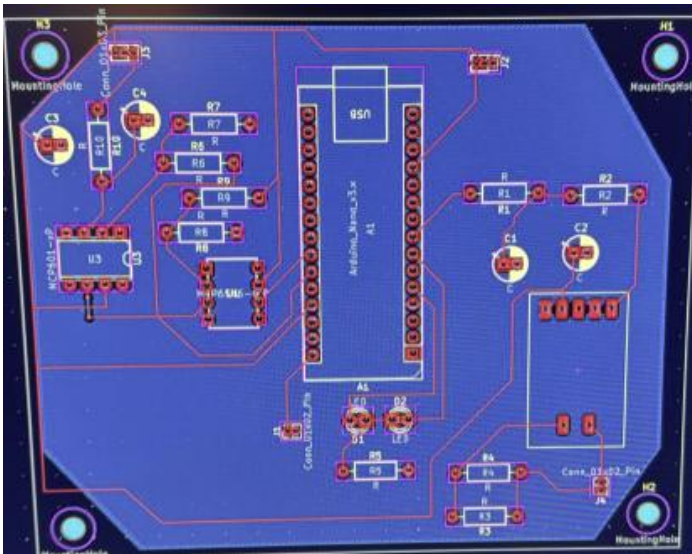


Figure 7: Top PCB (left) and Bottom PCB (right)

2. System Architecture

The ODRS system architecture is comprised of five main modules: the communication module, the vertical alignment module, the rotational module, the data collection module, and the mechanical module. These five subsystems work interactively to deliver all necessary functionalities to the system.

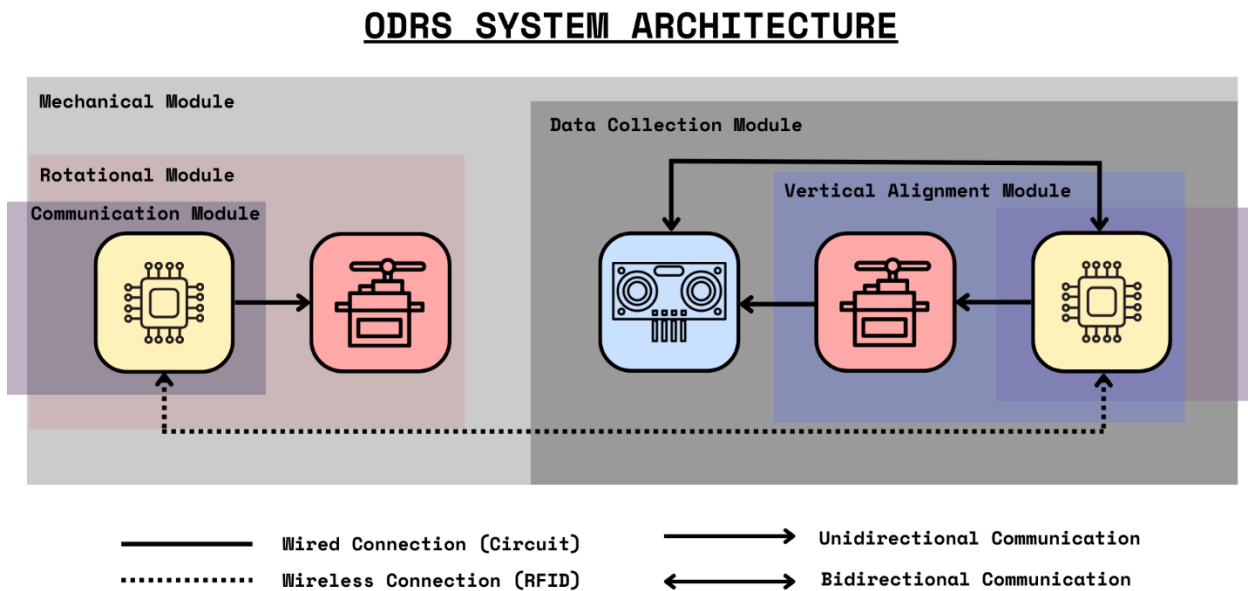


Figure 8: ODRS System Architecture Block Diagram

2.1 Module Description

Data Collection Module

This module is comprised of two ultrasonic sensors situated at opposite ends of the top platform, which are both mounted onto servos and controlled by the top microcontroller. This subsystem provides vision to the ODRS by continuously scanning for potentially disruptive objects. The range at which an object is deemed hazardous is chosen by the user – the sensor range can be toggled by inputting a new scanning range into the keypad matrix. Once a scan has occurred, this subsystem will provide the communication module with two values: a binary value (0 or 1) that indicates if an object has been detected, and the range at which it was detected (0 if nothing has been detected). User accessibility features are also incorporated into the system: red and green LEDs flash in correspondence with object detection, which provides users with real-time visual feedback.

Rotational Module

This module is comprised of a servo which spins the top platform in a continuous 360° rotation. Due to the high degree of interaction between subsystems, the rotational module must be meticulously synergized with the mechanical module to maintain the functionality of critical infrastructure. More information on these important design decisions can be found in [section 3.2](#). The goal of this module is to improve system reliability by reducing the likelihood of a blind spot, thus reducing the probability that the ODRS fails to detect an object.

Vertical Alignment Module

This module is comprised of two servos which can be rotated at precise angles based on PWM input from the top microcontroller (contrary to the servo in the rotational module). This module essentially acts like a subsystem to the data collection module, as there are many hardware components shared between modules (such as the two servos). However, this module has a different functionality, as it provides the ODRS with increased vertical detection range; to compensate for the ultrasonic sensor's ~30° useful range, the sensors can be rotated to a maximum of $\pm 45^\circ$, which quadruples the initial useful range by increasing it to ~120°.

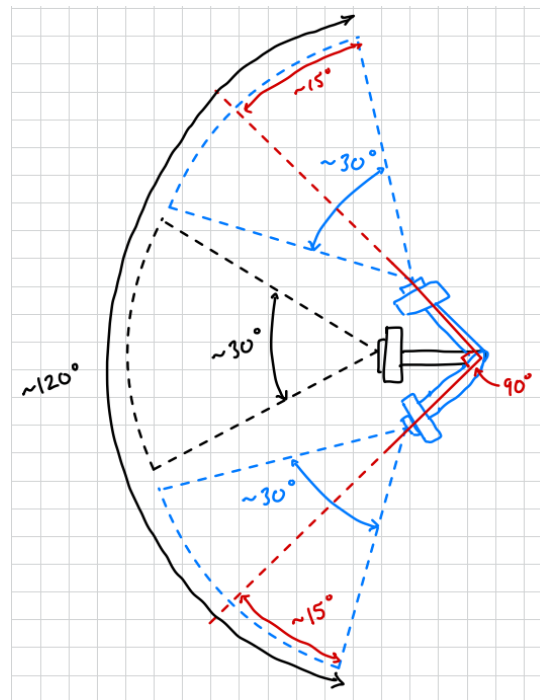


Figure 9: Ultrasonic Sensor Increased Useful Range Visual

Communication Module:

This module is comprised of the two microcontrollers, along with the circuitry that allows them to communicate with each other wirelessly. This RFID link was necessary to ensure the top part of the

system was able to safely rotate continuously and prevented any wires from coming undone, which would cause system failure. This module essentially serves as the backbone of the system, interconnecting all the modules together and ensuring full system integration. Furthermore, to allow the system to accommodate user input, the wireless link between microcontrollers must be a bidirectional communication channel – this will allow both data to be sent from the sensors to the user, and range/angle updates to be sent from the user to the sensors.

Mechanical Module

This module is comprised of various 3D printed components that are custom made to fit the needs of the ODRS. The module follows a four-layer design, with each layer serving a unique but essential purpose to ensure that all system functionalities are met. The bottom layer houses one of two PCB's, which includes the transmitter & receiver circuits, the bottom microcontroller, LEDs, and the wiring for the continuous rotation servo. The microcontroller on this layer also has USART capabilities to allow for user interfacing. The second layer is situated a few cm above the first layer and is designed to hold the continuous rotation servo – a layer specific to the servo was needed so that it could be centered inside the case without obstructing the PCB or wires underneath it. This layer also has special features to allow for the coil and Hall Effect sensor to be as centered as possible, which will increase the communication range by decreasing the distance between these components and the axis of rotation. A custom fitting then fits into the servo axle, which connects to the two other layers. The third layer houses the second PCB, which includes the transmitter & receiver circuits, the top microcontroller, LEDs, and the wiring for the two servos and ultrasonic sensors. Similarly to the second layer, the fourth layer is needed to prevent the servos/sensors from obstructing the PCB wiring, and includes 3D printed supports to ensure the servos are anchored securely. A more detailed view at the design process for this system can be found in [section 3.5](#).

2.2 Hardware Overview

The ODRS consists of two main PCBs which communicate wirelessly between each other via RFID communication. The hardware/circuitry components required for each PCB is as follows:

Top PCB:

- 2X Ultrasonic Sensors
- 2X Angle-precise Servos
- 4X LEDs
- Transmitter & Receiver Circuitry (including coil and HE sensor)

Bottom PCB:

- 1X Continuous rotation Servo
- 2X LEDs
- Transmitter & Receiver Circuitry (including coil and HE sensor)

Each module is interconnected through the hardware, which is representative of the high levels of system integration needed for successful functionality. The transfer of information begins at the sensor level with ultrasonic detection, which is then passed to the top microcontroller and deciphered between a detection or no detection (is the object inside or outside the range). Then, this information is processed on the software side into an 8-bit string that is converted to a sine wave, amplified and transmitted serially through the RFID link to the receiver side, where this new weaker signal is re-amplified and converted into TTL logic for software use. Structurally, the communication module is the main canal of information, which intakes smaller streams of information from other modules (such as the data collection module or the user input from the vertical alignment module).

2.3 Software Overview

The ODRS software components primarily consist of two different types: data transfer and data collection. The distinction between these two types depends on the subsystem in question:

Data Transfer Software

- Transmitter Program (Communication Module)
 - Converting PWM waves to Sine waves.
 - Shifting the wave frequency based on the binary digit being sent (FSK).
- Receiver Program (Communication Module)
 - Converting the number of “HIGH” pulses in a set transmission period into binary values.
 - Converting these binary values into integers or other more useful data types.

Data Collection Software

- Object Detection Program (Data Collection Module)
 - Convert ultrasonic pulse emission/recognition into a Boolean detection value and float range value.
- User Input Program (Vertical Alignment Module/Data Collection Module)
 - Convert user input from the keypad matrix into values that can be used to adjust both the servo angles and the range of the sensors.

3. Detailed Design

This section of the report discusses the complex thought process and calculations behind the development of each individual module during the design process.

3.1 Data Collection Module

The Data Collection Module serves as the eyes of the ODRS, enabling accurate and continuous monitoring of objects in proximity to the system. This module utilizes ultrasonic sensors,

specifically the HC-SR04, which operate by transmitting a high-frequency sound wave via the TRIG pin and listening for its echo through the ECHO pin after it rebounds off a nearby object.

Upon startup, the Object Detection Radar System (ODRS) initializes all system parameters and sensor readings to default values, ensuring a clean baseline for operation. Each sensor module is then engaged through a dedicated function, `get_distance()`, which triggers a measurement cycle to determine the distance between the ultrasonic sensor and any potential object within its detection field. The system enters a continuous scanning loop, actively polling both sensors in real time to detect objects within the predefined threshold ranges. This enables persistent monitoring of the environment and ensures the system can respond instantly to changing conditions.

To initiate a distance measurement, the microcontroller sends a 10-microsecond HIGH pulse to the TRIG pin. This triggers the sensor to emit an 8-cycle ultrasonic burst at 40 kHz. Once the pulse bounces off an object and returns, the ECHO pin goes HIGH for a duration proportional to the time it took for the wave to travel to the object and back. This time interval is captured using an internal timer on the microcontroller, ensuring high precision in the measurement process. The distance to the object is calculated using the formula:

$$\text{Distance (cm)} = \text{Time} \times C / 2 = T \times 0.0343 / 2$$

Where:

- T is the duration of the echo pulse in microseconds,
- $0.0343 \text{ cm}/\mu\text{s}$ is the speed of sound in air at room temperature (343 m/s),
- Division by 2 accounts for the round-trip of the ultrasonic pulse.
- C is the speed of sound (343m/s)

Once calculated, this distance is continuously updated and fed into the logic control system. The data is used to determine if an object falls within critical proximity thresholds: over 10 cm (safe), between 1–10 cm (warning), or under 1 cm (danger). These zones directly influence system behavior such as LED signaling, ensuring the radar system responds appropriately in real-time. This module is optimized for responsiveness and reliability, enabling the ODRS to function as an efficient and robust proximity alert system.

A dynamic feedback mechanism is implemented using two red LEDs (RED1 and RED2), which serve as proximity indicators. Their flashing frequency increases proportionally as an object approaches closer to the sensor, providing a clear visual representation of urgency. The flashing delay is governed by the relationship:

$$\text{Delay_ms} = 50 * \text{Distance(cm)}$$

As the distance decreases, the delay shortens, causing the LED to flash more rapidly. When an object is detected at a distance less than or equal to 1 cm, the corresponding red LED remains solidly ON to signal immediate proximity and potential hazard. Conversely, if no object is detected or the object is beyond 10 cm, the green LED remains ON to indicate a safe zone. This operational logic ensures

intuitive and responsive system behavior, making the ODRS suitable for real-time object monitoring and proximity alert applications.

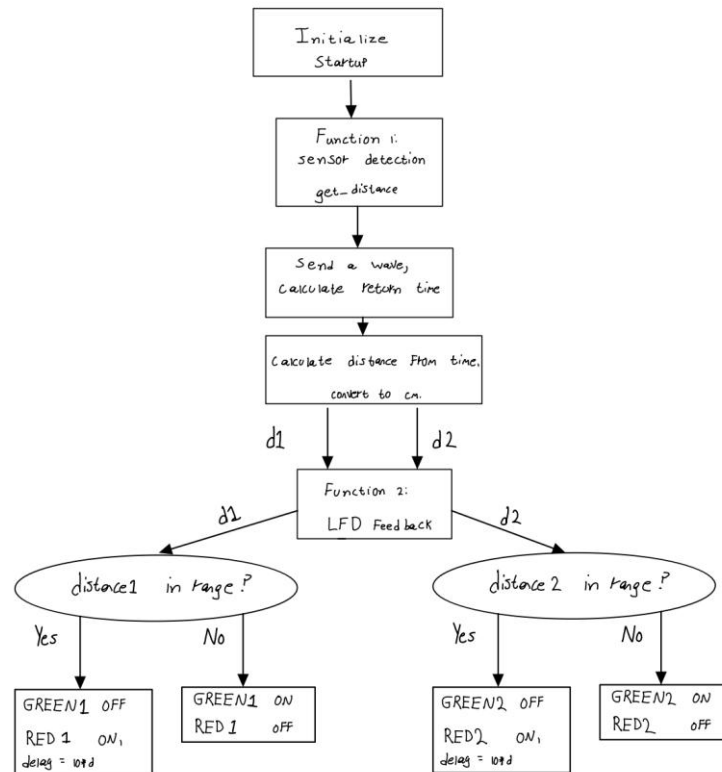


Figure 10: Data Collection Module Software Flowchart

3.2 Rotational Module

The Rotational Module is a core subsystem of the ODRS, designed to enable continuous 360° horizontal rotation for comprehensive environmental scanning. Its primary function is to minimize blind spots and maintain consistent object detection coverage as the system operates. Enables the rotation of the Data Collection Module, ensuring the ultrasonic sensors can scan the full surrounding environment. It achieves this by rotating the entire sensor platform, mimicking the behavior of conventional radar systems. The rotation is managed in real-time and is optimized to work passively without requiring interaction with other subsystems. The hardware for this module consists of:

- MG90S 360° Continuous Rotation Servo Motor
- Support Platform that mounts the ultrasonic sensors and rotates as a single unit with the servo shaft.

To spin the servo continuously, the microcontroller's 16-bit timer (Timer 1) was required, because the PWM frequency was quite low (only 50 Hz). This meant that the timer had to count to a value higher than 255 (max count value for an 8-bit timer) even with the highest pre-scaler setting (1024). A duty cycle of 7.5 would result in zero movement, and through trial-and-error, it was discovered that a duty cycle of ~7.8 was the best value to ensure the rotation of the 3rd and 4th layer maintained a stability at the correct speed (~20-30 RPM).

3.3 Vertical Alignment Module

The Vertical Alignment Module is used to adjust the vertical scanning angle of the two ultrasonic sensors attached to servo motors using a custom printed attachment. This subsystem allows the ODRS to operate its sensors reliably at various angles. Without this module, the ODRS range would be significantly reduced, due to the poor useful range of the ultrasonic sensors.

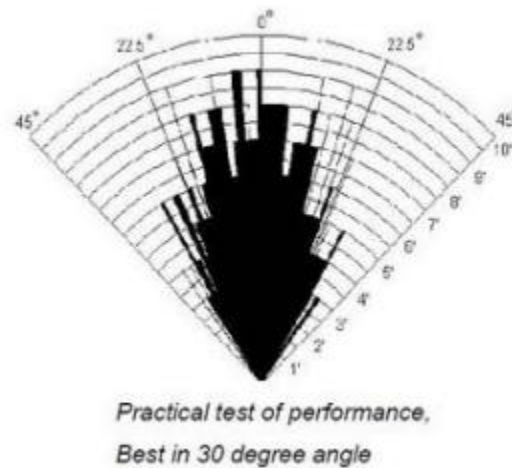


Figure 11: HC-SR04 Performance Test

The hardware for this module consists of:

- Two SG90 180° servo motors connected to pins PB3 and PB4.
- A 4x3 matrix keypad connected to: (rows: PC0-PC3, columns: PC4-PC5 and PD2).

The SG90 works by receiving a PWM signal with a pulse width between 1.0 ms and 2.0 ms, corresponding to a 90° and -90° respectively from the horizontal axis, so the center would require a 1.5 ms pulse width, which translates to 0°. Users input desired angles through the keypad, with * to reset, digits to enter the angle and # to confirm. However, due to hardware limitations and pin assignments used by other modules, the microcontroller's PWM pins were unavailable. Instead, a software generated PWM signal was implemented using Timer1 in CTC mode. Here is the methodology behind this method:

- Timer fires an interrupt every 10 μ s.

- TIMER1_COMPA_vect simulates the PWM waveform:
 - At tick 1: Servo 1 pin goes HIGH.
 - At tick = 1 + servo1_ticks: Servo 1 pin goes LOW.
 - At tick 1000: Servo 2 pin goes HIGH.
 - At tick = 1000 + servo2_ticks: Servo 2 pin goes LOW.
- servo1_ticks and servo2_ticks range from 100–200, corresponding to 1.0–2.0 ms pulse widths.

To begin, the user presses * to enter input mode. They can then enter a desired angle between $\pm 45^\circ$, followed by # to confirm the input. Positive angles are entered normally (e.g. 1, 5, # for 15°), while negative angles are entered using a 0 prefix which acts as a flag to indicate a negative angle. E.g. the inputs 0, 3, 0, # will be interpreted as -30° . Both servos can be reset to 0° at any point by pressing *.

Once the angle is confirmed, the program converts the string input into an integer using atoi() and checks whether the value is within range. Any angle greater than $\pm 45^\circ$ is ignored to ensure that the ultrasonic sensor does not hit and interfere with any other part of the device. If the angle is within the range, it will get scaled to a PWM offset relative to the servo's center position, which is 150 ticks. To calculate the appropriate number of the offset ticks for any given input angle, it is calculated using:

$$\text{offset} = (\text{magnitude} * 25) / 45$$

Here, magnitude is the absolute value of the user's entered angle, this offset is either added or subtracted, depending on the angle's sign, from the neutral 1.5 ms position (150 ticks) to produce the final pulse width.

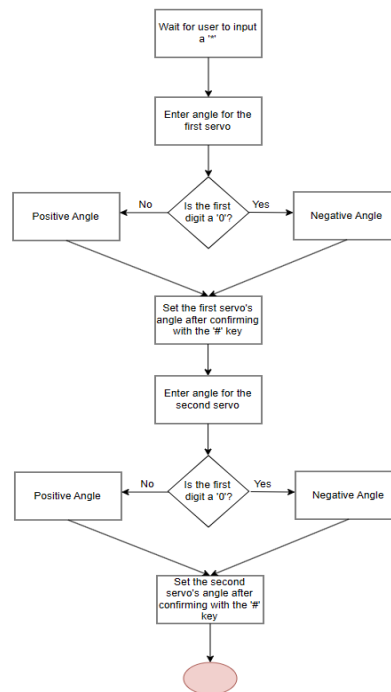


Figure 12: Vertical Alignment Module Software Flowchart

For example:

- $15^\circ \rightarrow \text{offset} = (15 \times 25) / 45 = 8$ ticks from the center
- $30^\circ \rightarrow \text{offset} = 16$ ticks from the center
- $45^\circ \rightarrow \text{offset} = 25$ ticks away from the center (maximum allowed)

This ensures that the user input angles are scaled correctly and linearly and that the resulted PWM pulse width stays within the set limit by rejecting any angles beyond $\pm 45^\circ$.

3.4 Communication Module

The communication module serves as the backbone of the ODRS, interconnecting its various modules together to ensure proper system integration. The primary objective of this module is to allow the two microcontrollers to communicate wirelessly in order to avoid impeding with other critical functionalities of the system. Furthermore, this communication link must be bidirectional to allow for important user customization features, which are needed to fully satisfy the problem definition. The importance of this module and its stringent requirements have resulted in a complex system that can be broken down into two parts: transmission and reception.

3.4.1 Transmitter

The method of communication being used in this system is known as frequency shift keying (FSK), which consists of transmitting information through sine waves by modifying the frequency of the wave. The communication module transmits data at ~ 140 bps (bits per second) at its slowest, with the 140 Hz frequency denoting a 0 and the 280 Hz frequency denoting a 1.

Transmission Software:

To transmit a signal effectively using RFID transmission, first a sine wave had to be generated from one of the I/O pins on the microcontroller. Unfortunately, the Arduino Nano clone that was used for the microcontroller did not support sine wave generation as a built-in feature, so this part of the transmission required some extra software to achieve successful implementation. To achieve this, PWM waves would be converted into sine waves in a two-stage technical process. The first stage (software) consisted of generating a PWM wave with an average DC value that mimicked that of a sine wave. When done correctly, each average DC value generated would represent a point along the desired sine wave – as this was only an approximation, the resolution of the wave could be modified by increasing or decreasing the amount of sampling points. To achieve this, a sine lookup table was created, which would contain the necessary duty cycle values in a sequential manner. The table creation process is as follows:

1. Choose number of sampling points
 - From experience, larger number of points led to smaller maximum wave frequency

- After some trial and error, it was found that 64 points was able to produce a clean wave while also maintaining adequately high frequencies, which were needed to maintain a data transmission rate of ~4 bytes/second.
 - Although not excessively quick, this data transmission rate would still be relatively quick, as it would update the user feedback 4 times per second.
- 2. Find the appropriate sine values
 - $360^\circ/64 = \# \text{ of degrees/division (deg/div)}$
 - Find the sine value V_N at each sampling point
 - $V_N = \text{sine}(0+N*\text{deg/div})$
 - Scale the wave to 5 V_{PP} and center wave at the origin (μc cannot output negative voltage values – add 2.5V DC offset)
 - $V_N = 2.5*V_N + 2.5 = V_{DC, AVG}$
- 3. Convert sine values into duty cycle percentages
 - These values represent the average DC value needed from the PWM
 - $V_N = V_{MAX} * \text{duty_cycle} \rightarrow \text{duty_cycle} = V_N / V_{MAX} = V_N / 5$
- 4. Adjust PWM values in real-time
 - Change the timer TOP value to switch between frequencies
 - Make PWM compare match value a function of duty cycle and the timer TOP value for seamless transition of duty cycle between frequencies (maintain wave generation for different frequencies)
 - Compare match value = duty_cycle * TOP

Now that a sine wave can be generated with varying frequencies, the last step is to send specific binary values that can carry information. To inform the system if a 0 or 1 must be sent, a binary reference value is used to pick out and compare specific bits from our desired number. This is done by shifting a one into a position N of the reference variable (which is otherwise all 0's) and then ANDing the reference variable and the number being sent. Any number ANDed with a 0 is 0, so the result of our operation will be a zero for every bit besides the bit in index N, which is a 1. Therefore, if we want to send a 0, the entire number will become a 0; otherwise, a 1 must be sent. An example of this critical operation:

$$\begin{array}{r}
 0b00000100 \\
 \& 0b10101101 \\
 \hline
 0b00000100
 \end{array}$$

In this example, the bit in position 2 ($N=2$) was revealed to be a 1, because the answer of our operation was anything other than zero (the answer will be 0b00000000 if the bit being sent is zero). Once the system knows if it is sending a 0 or 1, the frequency (and thus the duty cycle as well) are adjusted accordingly. Furthermore, there is another counter which keeps track of the number of wave periods sent per bit, where N periods sent per bit = $N*\text{sine_lookup_table_size} = N*64$. This allows the number of periods (or cycles) sent per bit to be adjusted easily, which was useful for trial-and-error testing on the receiver end. To keep the period of transmission per bit constant, 4 periods were sent for a 0 and 8 periods were sent for a 1.

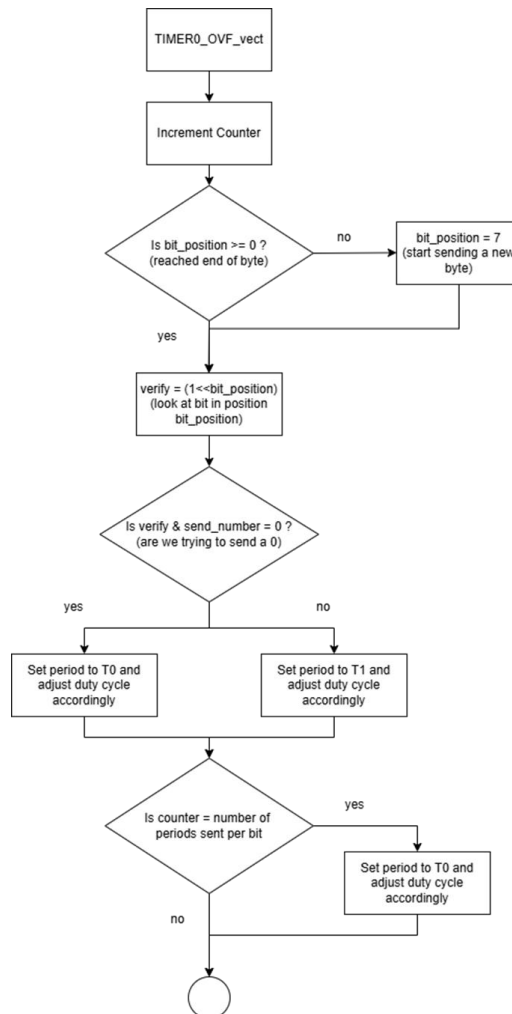


Figure 13: Transmitter Software Flowchart

Transmission Hardware:

The second stage of the conversion from PWM to sine waves took place through the transmitter hardware; once the correct time varying duty cycle was applied to the PWM waves, they had to be passed through a low pass filter to produce a clean sine wave. This is because the low-pass filter would filter out the wave's upper harmonics, leaving only the average DC value (which is what we want). The reason this works is because according to the Fourier Series, the PWM square wave is made up of a sum of many smaller harmonic waves. The term in the series with 0 frequency represents the average of the function (or average DC value in this case). Therefore, by filtering out the higher frequencies, the filter essentially extracts (at least approximately) the desired voltage values that were calculated in the sine look-up table. One problem encountered by the filter was that the higher frequency parts of the sine wave were being attenuated significantly more than the lower frequency parts. This could have been due to an unnecessarily elevated cutoff frequency – to solve the problem, a cutoff frequency of ~241 Hz was used, since it was high enough to have a small effect on the higher frequency of 280 Hz, but also low enough that the signal remained smooth. This value was calculated based on the following formula:

$$f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi(1000)(0.66 \cdot 10^{-6})} \approx 241 \text{ Hz}$$

After the filter, the signal is amplified using the Audio Amplifier, which will amplify the power of the signal, thus amplifying both voltage and current. This is done to drive more amperage through the coil, which will increase both the strength of the resulting magnetic field and the range of the device. The 8Ω power resistors were put in parallel to reduce their equivalent resistance, which should further maximize current flow.

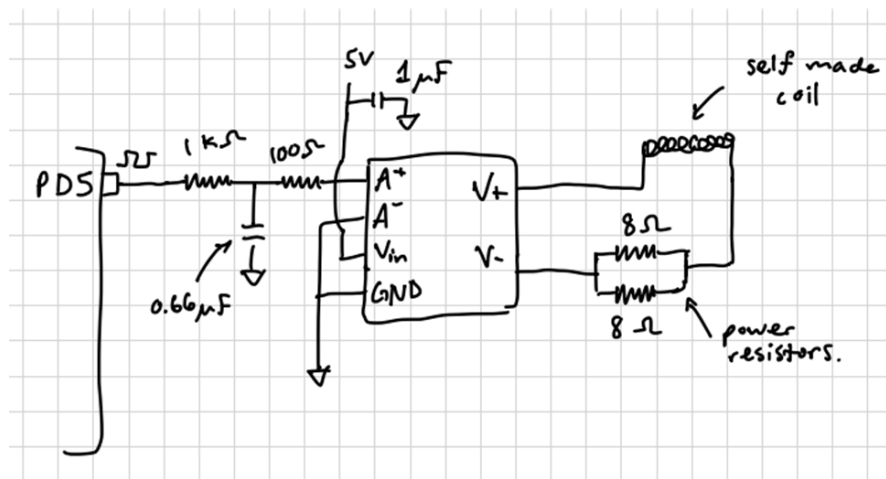


Figure 14: Transmitter Hardware Schematic

3.4.2 Receiver

At the receiver end, the hall effect sensor will ideally pick up the changing magnetic field and output an equivalent sine wave on the other end, which can then be converted to digital logic using a comparator. When the transmitter is not in range, the receiver side should read zero (noise should not cause false signal detections).

Receiver Software:

Once converted to digital logic by the comparator, the sine wave passed by the transmitter will be turned into a high or low signal, with the duty cycle based on how long the peak of the sine wave exceeded the reference voltage. To decipher the frequency of the input wave, the microcontroller's built-in input capture mode (ICR) was used. This mode allows an interrupt to be set every time the input capture pin (PINB0) detects a rising/falling edge at the pin. In this case, the settings were

configured so that the interrupt was only set for each rising edge, which would then increment a counter. The benefit of making the number of periods sent per bit easily adjustable was so that the period of transmission could be kept consistent through different frequencies. Since 4 periods are sent per 0 and 8 are sent per 1, the period of transmission per bit is:

$$T = \frac{1}{280} (8) = \frac{1}{140} (4) = \frac{1}{35} \approx 0.0285 \text{ s}$$

Therefore, the number of times the counter is incremented during the transmission period of 28.5 ms should indicate whether a 0 or a 1 was transmitted. Once this time is up, the counter is reinitialized to zero and the process restarts at the next rising edge detected (a new byte is being transmitted). The answer can then be stored in an array to be easily printed out or converted into an integer value. The program flowchart is as follows:

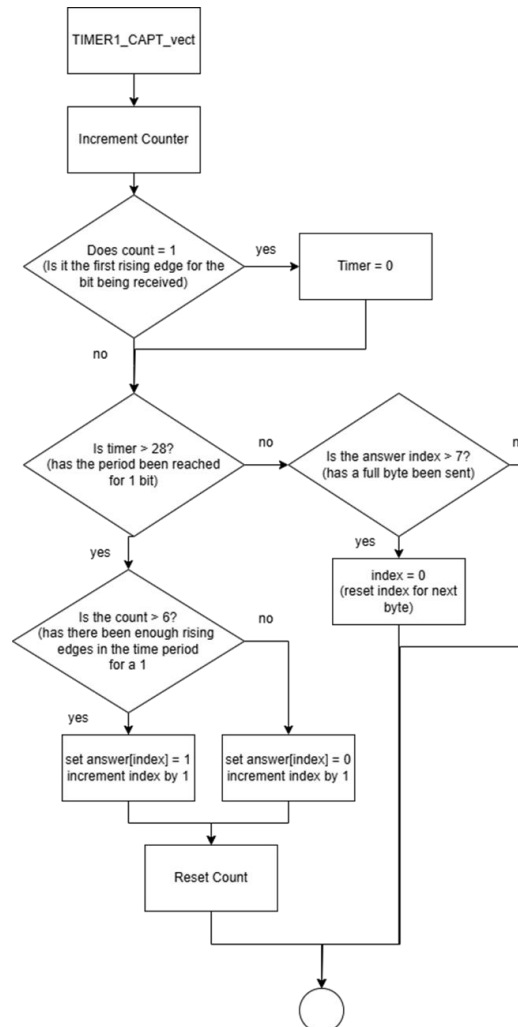


Figure 15: Receiver Software Flowchart

Receiver Hardware:

The receiver hardware served as the bridge between the transmission side and the receiver software. This critical circuitry had 3 main stages: signal detection, signal amplification, and signal logic conversion. In the signal detection stage, the changing magnetic field caused by the sine waves passing through the coil is picked up by the hall effect sensor and translated into an output voltage on the receiver side. This output signal can become very noisy, so it is passed through a low pass filter with a cutoff frequency of ~241 Hz (again). This will produce a clean signal on the other side which is transmitted wirelessly through magnetic fields.

The second stage is the signal amplification stage, where the very weak signal coming out of the low-pass filter gets amplified for use in the final stage. Firstly, because the hall effect sensor outputs a signal with a slight 1-2V DC offset, a coupling capacitor is added to the input of the op amp (non-inverting terminal in this case) to remove that offset. This is done to avoid clipping, where the signal is amplified higher than the op amp's $+V_{CC}$ value, thus saturating the op amps gain capabilities (no longer in its linear operational state). Once the offset is removed, the very weak hall effect signal is amplified with a gain of ~375 to achieve a signal large enough to be differentiable to noise while still being identifiable by the comparator. The op amp gain is calculated using the formula:

$$Gain = 1 + \frac{R_f}{R_{in}} = 1 + \frac{56000}{150} \approx 375$$

Finally, the logic conversion stage aims to convert the analog signal into digital logic using a comparator. The reference voltage of the comparator was set at the negative comparator pin, which means that it would output 5 volts if the signal at the positive pin was higher, and 0 if it was smaller. The reference voltage was initially set at around 0.25 V – while this is very low, it gave the transmitter a significant range boost. However, this seemingly massive range was an illusion; the low reference voltage combined with the large signal amplification meant that general noise was getting amplified beyond the threshold of the reference voltage, meaning the system was getting false signals. This was fixed by adjusting the voltage value using trial-and-error experiments. Here it was revealed that the system yielded the satisfactory results when the reference voltage was set to ~1.67 V. This was done using a voltage divider that fed into the comparator's negative pin – let the resistor to ground be R1, and the resistor to 5V be R2.

$$V_{REF} = 5 \cdot \frac{R1}{R1 + R2} = 5 \left(\frac{100}{100 + 200} \right) = 1.67 \text{ V}$$

Finally, to activate the BJT in the comparator's final stage, a pull-up resistor from the comparator output to 5V must be added (as shown in figure 16). This is then fed into the ICP pin to be interpreted into binary logic by the software.

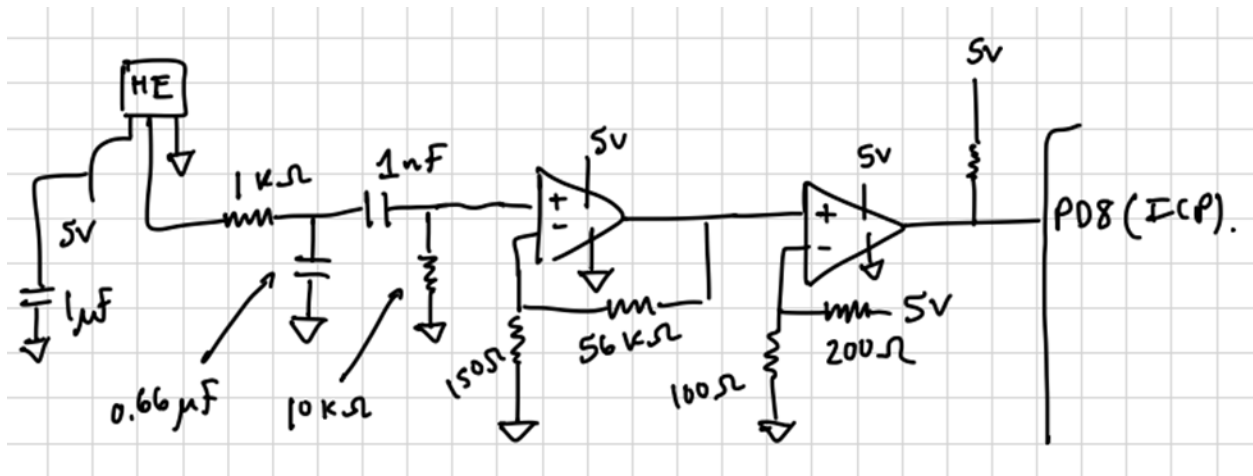


Figure 16: Receiver Hardware Schematic

3.5 Mechanical Module

The Mechanical Module serves as the physical framework that supports and integrates all the different components. It is designed to be compact and modular, giving easy access to any section of the ODRS.

The module consists of five primary parts:

i. Layer 2

Layer 2 is where the 360° MG90S servo is anchored, this layer is static, which allows the top parts to spin with the servo. The first layer is also used to hold the lower hall effect sensor and iron rod, which is essential as we need them to be static to not tangle when the device rotates.

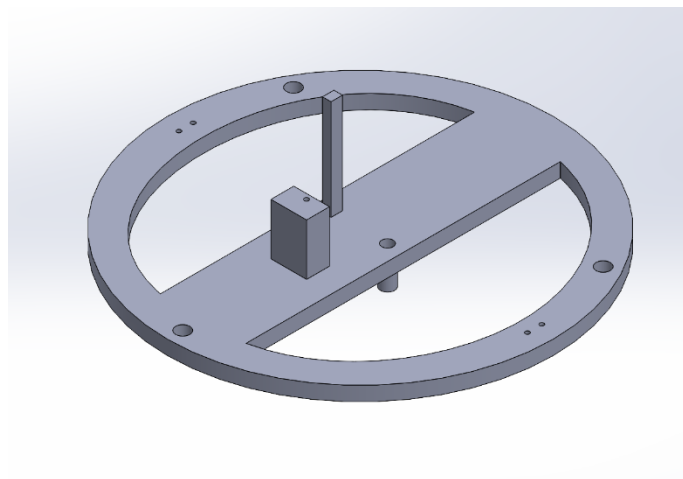


Figure 17: Layer 2 CAD Model

ii. Layer 3

Layer 3 is mounted directly on the shaft of the MG90S servo and is designed to rotate with it. This layer carries the upper PCB (which was switched to a mini breadboard) that has the connections to the vertical alignment module/data collection module (two HC-SR04 sensors and two SG90 servos) alongside the top hall effect sensor and coil.

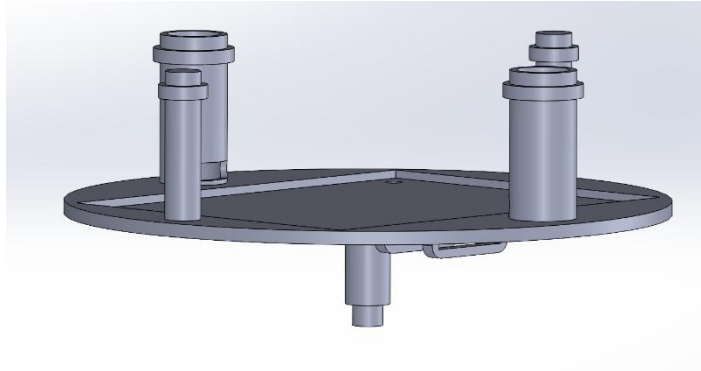


Figure 18: Layer 3 CAD Model

iii. Layer 4

Layer 4 is the top platform, which houses both the vertical alignment and data collection modules, as well as a battery pack that is used to power the top microcontroller.

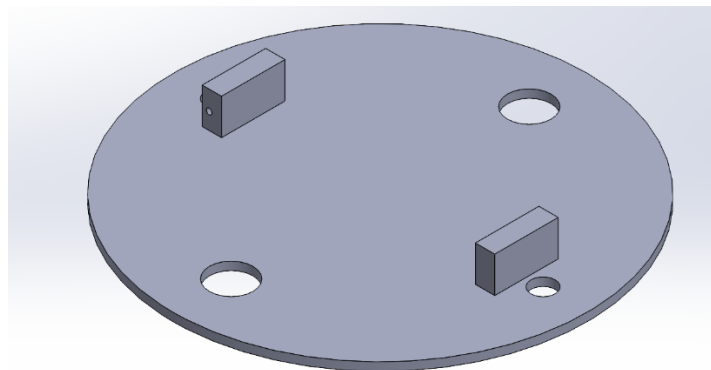


Figure 19: Layer 4 CAD Model

iv. Base/Layer 1

The base houses all parts and components, which was essential to give the ODRS a complete product look to it. The base had to be split into two attachable parts, otherwise it would have taken over 19 hours to print, so we had to divide the base to facilitate the printing process.

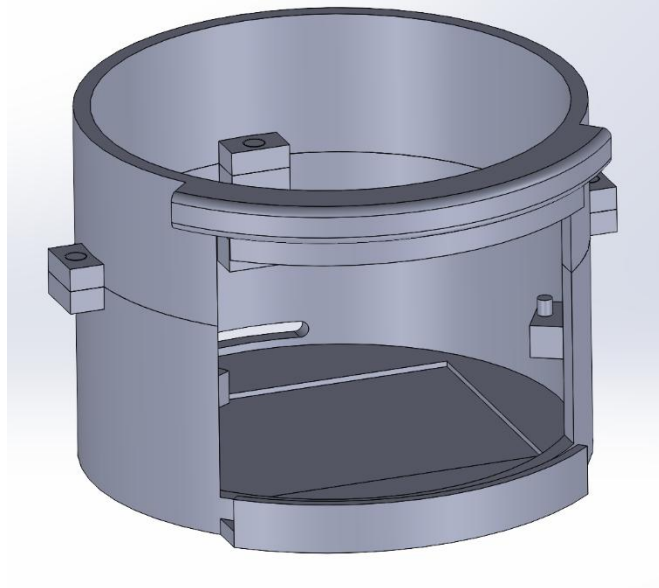


Figure 20: Base Section A & B CAD Models

v. Door

The sliding door provides quick and easy user access to internal wiring and components for quick adjustments and fixes. See [Exploded View](#) for more information.

4. System Performance

This section discusses the performance of the finalized system on a module-by-module basis, with respect to the initial functional requirements that the system was expected to deliver.

4.1 Module Testing

Both during and outside the demo, many tests were completed to make sure the ODRS was fully capable of all required functionalities. When it was finally time for the demo, the following tests were performed, as they were deemed most representative of complete system functionality.

4.1.1 Data Collection Module

To verify the reliability and responsiveness of the **Data Collection Module**, a series of tests were conducted to evaluate object detection accuracy and real-time feedback consistency. The tests focused on two primary conditions: detecting static objects within a defined sensor range and tracking dynamic objects moving within the radius of both ultrasonic sensors. The results are summarized in *Table 1*, confirming that the module performs as expected under both conditions, providing dependable environmental data and accurate feedback via LED indicators.

Table 1: Data Collection Module Test Results

<i>Test</i>	<i>Expected Result</i>	<i>Pass/Fail</i>	<i>Conclusion</i>
<i>Placing object within set range of individual sensors</i>	GREEN before object enters range, flashing red at increasing speeds when object is within range	PASS: Sensors were able to accurately detect an object within its set range, consistently, at different ranges.	Any object placed in front of the sensor will consistently toggle the LED feedback system, for any set range.
<i>Moving object within radius of both sensors</i>	Both sensors should control only their lights and have no noticeable delay in sensing moving objects.	PASS: Both sensors were able to accurately detect moving objects with varying distances, consistently.	A moving object can be sensed in real time by both sensors, even with varying distances from the sensors.

4.1.2 Rotational Module

The **Rotational Module** was evaluated for its ability to maintain smooth, continuous rotation and ensure reliable data acquisition from the ultrasonic sensors during movement. The following tests verify its mechanical stability and operational synchronization with the Data Collection Module.

Table 2: Rotational Module Test Results

<i>Test</i>	<i>Expected Result</i>	<i>Pass/Fail</i>	<i>Conclusion</i>
<i>Continuous 360° rotation under load</i>	Servo rotates platform smoothly at ~20-30 RPM without stalling or jittering	MARGINAL PASS: Platform occasionally rotated continuously and stably under full system weight, however the friction from the wires sometimes caused rotational issues.	Optimizing wire placement using electrical tape or other measures would ensure a more smooth and constant rotation.
<i>Synchronization with sensor scan delay</i>	Rotation speed allows ultrasonic sensors to detect return signals without error	PASS: Servo speed was tuned to ensure accurate readings without missing echo returns.	The servo speed was not inhibiting sensor accuracy.

4.1.3 Vertical Alignment Module

The **Vertical Alignment Module** was tested to determine its responsiveness and accuracy when converting the user input into an accurate servo angle. The following table summarizes the performed tests, describing the expected behaviour, results, and further observations.

Table 3: Vertical Alignment Module Test Results

<i>Test</i>	<i>Expected Result</i>	<i>Pass/Fail</i>	<i>Conclusion</i>
<i>Control servo angle manually using the keypad matrix</i>	Servo 1 will rotate at the angle first input by the user, then the second input will rotate Servo 2. The servos should be accurate within $\pm 2.5^\circ$.	PASS - Both Servo 1 and Servo 2 react to keypad input and successfully change their angle within a margin of $\pm 2.5^\circ$.	The servos responded accurately to user input; the module is fully functional and ready for integration.
<i>Reject angles outside valid range (e.g. $\pm 50^\circ$)</i>	Inputs beyond $\pm 45^\circ$ should be ignored. No movement should occur.	PASS - Out of bound inputs were rejected as expected, no signal was sent to the servos.	The module correctly filters invalid input and prevents larger angles.
<i>Negative angle input via the 0-prefix format (e.g. '0 3 0 #')</i>	System interprets the input as -30° and adjusts Servo 1 or 2 accordingly.	PASS - Negative inputs are correctly converted to valid servo angles.	Negative angles are handled reliably, the input system is intuitive and consistent.

4.1.4 Communication Module

The **Communication Module** was developed to enable real-time, wireless transmission of user inputs and sensor data between the rotating and static components of the ODRS. To assess its functionality, tests were conducted to evaluate dynamic servo and sensor range adjustments during system operation. While the standalone performance of the module was functional, integration issues prevented full system synchronization. **Table 2** outlines the testing outcomes, highlighting the module's individual capability and the integration challenges encountered

Table 4: Communication Module Test Results

<i>Test</i>	<i>Expected Result</i>	<i>Pass/Fail</i>	<i>Conclusion</i>
<i>Real time sensor range adjustment based on user input to the keypad matrix</i>	After adjusting the sensor range using the keypad, the sensors would adjust in real time while the device was still spinning.	FAIL: The RFID link was not fully integrated with the system, so the device failed at implementing real time range updates.	While the module worked on its own, it had not been integrated with the other modules in time, resulting in system failure.
<i>Real time servo angle adjustment based on user input to the keypad matrix</i>	After adjusting the servo angles using the keypad, the servo angles would adjust in real time while the device was still spinning.	FAIL: The RFID link was not fully integrated with the system, so the device failed at implementing real time angle updates.	While the other modules worked on their own, the communication module failed to serve its purpose in fully integrating the system.

4.1.5 Mechanical Module

To verify the reliability, accessibility, and structural integration of the mechanical design, several physical tests were carried out on some of the major features of the module. That included rotational stability, modularity of the layered structure, and accessibility of the door. The table below is a summary of the tests and conclusions drawn from each test.

Table 5: Mechanical Module Test Results

<i>Test</i>	<i>Expected Result</i>	<i>Pass/Fail</i>	<i>Conclusions</i>
<i>Layer 2 - Servo connection stability</i>	360° servo securely attaches to Layer 2 for stable rotation.	PASS - The servo was able to rotate the top parts.	Although the top layers were able to spin, the connection loosened over time which made the spin wobble a bit.
<i>Sliding door access</i>	Door opens/closes smoothly for easy internal access.	FAIL - Modified base design blocked the sliding path.	The last-minute change to the base should have accounted for the sliding door.
<i>Layer fit and modularity</i>	All layers align and fit together without force.	PASS - All pieces were able to be assembled and disassembled easily.	Modular design works as intended and allows easy removal and access to any part for maintenance.

4.2 Demo Results

The results achieved in the demo were representative of the system's status:

Usefulness: 4/5

Innovation: 4/5

Feasibility: 2/5

While the project was recognized for its real-world applications and its innovative concepts and designs, the lack of complete system integration was a clear sign that the product would not yet be feasible to a real-world client. This is reflective of the test results, which showed that the modules mostly all worked individually at very high success rates, however when trying to integrate the various subsystems together, the task became quite complicated. Unfortunately, certain phases of project development exceeded their intended timelines (see [Gantt Chart](#)), which resulted in a lack of time for system integration, which was the final step in the development process.

5. Conclusion

This section of the report is a global synthesis of the entire project, discussing both the strengths and weaknesses of the project, the project innovations, points to improve on in hindsight, as well as potential applications of an ideal finished product.

5.1 Project Summary

The Object Detection Radar System (ODRS) represents a multidisciplinary engineering effort aimed at creating a modular, real-time object scanning system. Designed for continuous 360° awareness and configurable vertical detection, the system demonstrates the viability of combining object detection, servo control, and wireless communication into a compact and intelligent embedded system.

The ODRS features ultrasonic-based data collection, bidirectional communication between two microcontrollers, and a four-layer mechanical structure designed in SolidWorks for seamless integration. The vertical alignment module enhances the detection field through user-defined angle control, while the rotational module ensures panoramic coverage. Despite challenges, the system successfully proved the functionality of each module in isolation, laying the groundwork for a fully integrated autonomous radar solution.

Throughout the project, one of the most critical takeaways was the importance of prioritizing **system integration** over incremental enhancements. While considerable time was spent on tasks like soldering/desoldering PCBs, they ultimately did not contribute to final performance and diverted resources from testing and merging core system functionalities. Earlier and more frequent software integration could have avoided the complications of debugging large monolithic codebases at later stages.

Despite these setbacks, the project served as an **exceptional learning experience**. Every team member gained valuable experience in circuit design and testing, short-range wireless communication, SolidWorks CAD modeling, and real-time embedded programming using AVR microcontrollers. The iterative design process, from mechanical prototyping to debugging timing-sensitive code, built a deeper understanding of how engineering systems behave in real-world scenarios. If given the opportunity to revisit the design, several key improvements would be implemented:

- **Earlier System Integration:** Ensuring that all modules are merged and functional as early as possible would streamline debugging and highlight compatibility issues.
- **Modular and Reusable Wiring Design:** Reducing wire clutter and improving physical layout of breadboard circuits would ease maintenance and reduce chances of connection errors, which were faced consistently throughout the project.
- **Avoid Custom PCBs in Early Prototyping:** Due to PCB's ultimately being our downfall, relying on breadboards until the final integration phase would save time and would have resulted in an overall complete system.

Despite the challenges, the project achieved several standout accomplishments:

- Developed an **innovative concept** with strong real-world applicability in sectors like automation, robotics, and navigation.
- Designed and tested a **bidirectional short-range wireless communication link**, allowing dynamic user interaction and sensor control without physical tethers.
- Created a **fully functional mechanical structure** using SolidWorks, which synchronized seamlessly with all hardware modules.
- Demonstrated **independent functionality** of all core subsystems — including object detection, servo-driven scanning, user input parsing, and data transmission.
- Gained hands-on experience in **RF signaling**, PWM control, embedded C programming, and system-level hardware/software integration

5.2 Potential Applications

The Object Detection Radar System (ODRS), with its modular design and multi-directional sensing capabilities, presents a wide range of potential applications across commercial, research, and defense sectors. Its real-time object detection, configurable scanning range, and wireless communication make it adaptable for both indoor and outdoor environments where autonomous navigation and spatial awareness are essential.

1. Indoor Robotics and Smart Home Devices

The ODRS can be integrated into robotic vacuum systems like **Roomba**, enabling enhanced obstacle avoidance and intelligent pathfinding through 360° object scanning. With further

development, the system could support room mapping and adaptive cleaning based on environmental changes.

2. Autonomous Delivery Platforms

By incorporating motion tracking and long-range communication modules (e.g., Wi-Fi or LTE), the ODRS could serve as the foundation for **autonomous delivery robots** used by services such as **Uber Eats**. It would provide these platforms with reliable local object detection and dynamic path adjustment, ensuring safer navigation in urban environments.

3. Reconnaissance UAVs (Unmanned Aerial Vehicles)

In military or scientific research contexts, the ODRS could be adapted to lightweight **surface or aerial drones** for low-altitude reconnaissance missions. Its compact architecture and low-power sensors make it suitable for real-time environmental scanning in terrains where GPS or satellite mapping may be unreliable.

4. Industrial Automation and Safety Systems

The system can be implemented in **automated factory floors** or **warehouse robotics**, offering an added layer of proximity awareness to prevent collisions between autonomous machinery and human workers, or to trigger emergency stops when objects enter predefined danger zones.

5. Educational and Research Tool

With its fully open and modular design, the ODRS is also ideal as a **teaching platform** for students and researchers learning about embedded systems, signal processing, and robotics. Its integration of mechanical, electrical, and software components makes it a valuable demonstration of real-world engineering systems.

References

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Operational Amplifier Datasheet:

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Appendix A

